

CLINICAL TOXICOLOGY



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April 2025

Clinical Toxicology

Definition

- The discipline (within toxicology) concerned with the toxic effect of agents whose intent is to *diagnose, treat, or prevent* disease states in *humans*
- Also include agents used with non-therapeutic intent for example, alcohol and drugs of abuse
- Deals with diseases caused by toxicants

Prevention & treatment of poisoning

■ Goals of treatment

- To maintain the **vital** functions if their impairment is imminent
- To keep the **conc. of poison** in the crucial tissues as **low** as possible by
 - preventing absorption
 - enhancing elimination
- To **combat the pharmacological and toxicological effects** at the effector sites

Clinical management of toxicologic emergencies

Involves a stepwise approach

1. Stabilization of the patient...
2. Clinical evaluation (history, physical examination, and laboratory and/or radiological tests)
3. Prevention of further absorption, exposure, or distribution
4. Enhancement of elimination of the suspected toxin
5. Administration of an antidote
6. Supportive care and follow-up

Stabilization of the patient

- Involves:
 - General assessment of the situation & the environment
 - Overall appearance of the patient
 - Maintenance of vital signs
- Removal of the victim from the obvious source of contamination, such as from fumes, gas, or spilled liquid.
- Maintenance of the **ABCs** of clinical management

Clinical evaluation

- **History of the event**

- Determination of what **substance ingested**, dose
- **Time of exposure** (when)

- **Physical examination:** Identification of clinical signs and symptoms that are likely associated with exposure from certain classes of toxic agents.

- The identification of the toxic syndrome or **“toxidrome,”**

TABLE 3.5

Common Toxidromes and Their Clinical Features

Anticholinergic	Dry mucous membranes, flushed skin, urinary retention, decreased bowel sounds, altered mental status, dilated pupils, cycloplegia
Sympathomimetic	Psychomotor and physical agitation, hypertension, tachycardia, hyperpyrexia, diaphoresis, dilated pupils, tremors; seizures (if severe)
Cholinergic	SLUDGE (sialorrhea, lacrimation, urination, diaphoresis, gastric emptying), BBB (bradycardia, bronchorrhea, brochospasms); muscle weakness, intractable seizures
Opioid	CNS depression, miosis, respiratory depression, bradycardia, hypotension, coma
Benzodiazepine	Mild sedation, unresponsive or comatose with stable vital signs; transient hypotension, respiratory depression

History & Physical Examination

History

- Fact finding mission – patient, family, friends, GP, empty packets, vomit, anyone !!??
- What ?
- When ?
- How ?
- How much ? (mg / kg)
- What else ? (Including alcohol)
- Why ?
(to determine substance ingested or to know the time and extent of exposure)

Physical examination

- vital signs
- eyes and mouth
- skin
- abdomen
- nervous system

Laboratory Procedures

- Initializing emergency measures should not depend on obtaining laboratory results.
- Qualitative and quantitative analysis
- Arterial blood gases (*Normal* or $\uparrow AG$)
- Electrolytes
- Renal function tests
- Serum osmolality
- Electrocardiogram

Differential Diagnosis of Metabolic Acidosis with Elevated Anion Gap: "AT MUD PILES"

A	Alcohol (Ethanol ketoacidosis)
T	Toluene
M	Methanol
U	Uremia
D	Diabetic ketoacidosis
P	Paraldehyde
I	Iron, Isoniazid
L	Lactic Acid
E	Ethylene Glycol
S	Salicylate

Prevention of further absorption

- Emesis induction
- GI decontamination



Enhancement of Poison Elimination

- Alkalinization of the urine- for some weak acids.
- Hemodialysis,
- Hemofiltration,
- Multidose activated charcoal (MDAC)...Q 2-4hrs
- Plasma exchange or exchange transfusion

Drug toxicity

specific antidote

- | | |
|-----------------------------|----------------------|
| ● Atropine | Physostigmine |
| ● Benzodiazepines | Flumazenil |
| ● Opioids (e.g., morphine) | Naloxone |
| ● Paracetamol/Acetaminophen | N-acetylcysteine |
| ● Heparin | Protamine |
| ● Warfarin | Vitamin K |
| ● Digoxin | Digoxin-specific Fab |
| ● Iron | Deferoxamine |
| ● Isoniazid (INH) | Pyridoxine |
| ● Beta-blockers | Glucagon |
| ● Cyanide | Sodium thiosulfate |
| ● TCA | Sodium bicarbonate |
| ● Ethylene glycol | Fomepizole |
- Antidotes in the market?

Supportive care and follow-up

- Poisoned patients who are unstable or at risk for significant clinical instability are admitted to *ICU* for close monitoring
- Psychiatric assessment- intentional self poisoning, once the patient is stable

SPECIFIC POISONINGS

1. Sedative hypnotics

- Sedative-hypnotics can cause dose-related depression of the CNS – this is the basic cause of adverse effects
- Low doses - drowsiness, impaired judgment, diminished motor skills
- Overdoses (deliberate)- toxicities

Benzodiazepines

- ↑s the frequency of the opening of the *chloride* channels & facilitate the actions of *GABA*.
- At high doses, induce CNS dep, muscle relaxation and cause vasodilation & hypotension (after i.v. administration).
- Do not significantly alter ventilation, except in:
 - Patients with respiratory complications
 - Elderly population
 - Presence of alcohol or other S/H.

Benzodiazepines....

- **Overdose:** *ataxia, drowsiness*, sedation and motor incoordination, paranoia or erratic behavior,
- Mild respiratory depression & hypotension
- Mild CV depression
- Coma is rare

Benzodiazepines

- Clinical management is symptomatic, and may also incorporate the use of a specific antidote.
- *Flumazenil*, is a benzodiazepine antagonist.
- Flumazenil reverses the sedative, anxiolytic, anticonvulsant, ataxic, anesthetic
- Administration of 0.2 to 1.0 mg i.v. has an acute onset of 1 to 3 min and a peak effect at 6 to 10 min.

Barbiturates

- Barbiturate poisonings may be seen in intentional (suicidal) poisoning but less frequently encountered in accidental poisoning
- In 80s and 90s most suicides were related to barbiturate overdose (alone or with alcohol) but not now...



Signs and symptoms

- Related directly to CNS and cardiovascular depression.
- Reactions are dose-dependent and vary from mild sedation to stupor and coma
- At highest doses, blockade of sympathetic ganglia triggers hypotension, bradycardia & ↓ inotropy, with consequent ↓ CO.
- Respiratory acidosis results from accumulation of CO_2 , shifting pH balance to the formation of carbonic acid.
- Patient presents with hypoxic shock, cold and sweaty skin (hypothermia), and either slow, or rapid, shallow breathing
- Coma is common

❑ Management of barbiturate poisoning

Stabilization

- Patent airway- A clear airway is ensured by thorough suctioning
- Adequate **ventilation**
- Keeping the patient warm
- Cardiovascular support- *inotropic* agents

❑ Enhancing drug elimination & ↓g absorption

- Gastric lavage- If comes within 2-4 hrs of ingestion
- Activated charcoal- 1g/Kg-1 is administered through NGT; Cathartic can be used along with it for further removal of barbiturates
- Multidose activated charcoal (MDAC) increases the clearance and decreases the half-life of phenobarbital
- Forced alkaline diuresis: especially useful in long acting barbiturates which are largely excreted by the kidney (phenobarbital) (NaHCO_3 1-2 meq/kg every 4-6 hr)
- Hemodialysis- If renal or cardiac failure, electrolyte abnormalities, or acid-base disturbances occur

2. Anticholinergic and Neuroleptic Drugs

❖ Anticholinergic, antihistamines

- Block central & peripheral cholinergic receptors
- Dry mouth, dilated pupil & blurred vision: most common ADRs of all anticholinergics
- HTN, dyspnea, urinary retention, constipation, dry & flushed skin
- ↑ vascular permeability and capillary perfusion: unregulated SNS stimulation, and accounts for the facial and upper body flushing
- Sinus tachycardia is the most sensitive sign of toxicity
- Confusion/delirium, agitation...
- Sedation: common to antihistamines (blocking of central H₁ receptors)

Clinical management of acute overdose

- Anticholinergic poisoning: IV physostigmine, a reversible AchEI.
- Physostigmine reverses anticholinergic activity & ameliorates coma, delirium & seizures (act on both central & peripheral receptors)
- Physostigmine is not recommended in reversing the anticholinergic actions of tricyclic & phenothiazines due to potential induction cardiotoxicity, resulting in heart block/fatal asystoles.

❖ Tricyclic antidepressants (TCAs)

- Rapidly absorbed from GIT & first pass metabolism
- Highly protein bound & high v_d
- Longer half life of elimination > 24 hours
- Metabolised by hepatic enzymes
- Some metabolites active
- Conjugated and excreted by the kidneys

MOA of toxic effects of TCAs

1. Inhibition of NE reuptake nerve terminals
 2. Direct alpha adrenergic block-vasodilation
 3. Membrane stabilizing or quinidine-like effect on the myocardium
 4. Anticholinergic action
 5. H1
- CNS depression, seizures & cardiac arrhythmias are common.

Anticholinergic effects

- mydriasis, blurred vision, vasodilation (aggravating the hypotension), urinary retention, constipation & dyspnea (resulting from lower bronchial secretions)

CV effects

- Cardiac arrhythmia
- Quinidine-like depressant effects that cause slowed conduction with a wide QRS interval & depressed cardiac contractility
- Sinus tachycardia – most common (Chol. Blockade) but later bradycardia(quinidine like effect)
- Heart block
- Hypotension (α blockade)

CNS effects

- Drowsiness, sedation
- Convulsions
- Rigidity
- Delirium
- Respiratory depression
- Ophthalmoplegia
- Coma -decreased level of consciousness

Clinical management of acute overdose

- Monitor CV and respiratory functions (12 h up to 6 days).
- Ventilatory support in respiratory depression.
- Treatment with i.v. *fluids & vasopressors*, such as NE or phenylephrine: to reverse hypotension.
- Widening QRS complex: NaHCO₃ (provides a rapid ↑ in extracellular *sodium*)
- Resistant ventricular dysrhythmias: lidocaine
- Anticonvulsants for seizure management.
- *Physostigmine* is *not* recommended for TCA toxicity, due to potential induction of fatal asystoles.
- Hemodialysis: not useful because of the high V_d & high protein binding with TCAs.

❖ Antipsychotics

- FGA exert their actions by antagonizing dopamine receptors.
- Affect central DA pathways
 - **Nigrostriatal** tracts- abn motor movt (tremor, rigidity...)
 - **Mesolimbic**- hallucinations & delusions,
 - **Mesocortical** -ve psychosis symptoms...
 - **Tuberoinfundibular**- prolactin release
- affinities for *non dopaminergic* sites, such as cholinergic, α_1 -adrenergic & histaminic receptors: varied ADRs produced
- display sedation, muscle relaxation & lowering of seizure threshold
- depress the reticular activating system (RAS)

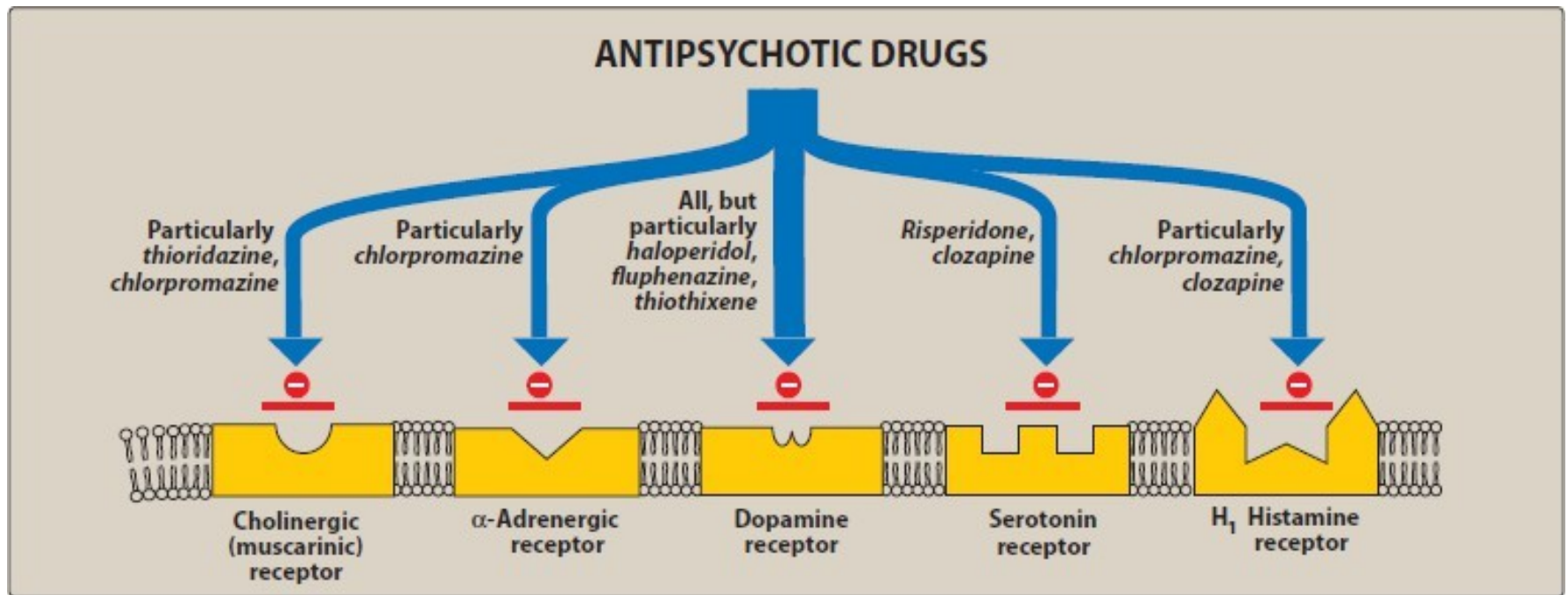


Figure 11.4

Antipsychotic drugs block at dopaminergic and serotonergic receptors as well as at adrenergic, cholinergic, and histamine-binding receptors.

Phenothiazines, some 2nd gen AP(clozapine)

- Have antimuscarinic & antihistaminic effects (such as mydriasis, dry mouth, tachycardia & ↓d GI activity).
- Vasodilation, orthostatic hypotension
 - α -adrenergic blockade contributes to hypotension.
- Quinidine-like CV AV block precipitates potentially fatal ventricular arrhythmias & cardiac arrest.
- EP signs and symptoms develop when an imbalance b/n antidopaminergic (greater) & anticholinergic activity is created.
- The more the antidopaminergic properties, the greater the severity of **EPS**(extrapyramidal symptom) reactions.
- EPS and NMS are *both ADRs* of neuroleptic drugs

Feature	Extrapyramidal Symptoms (EPS)	Neuroleptic Malignant Syndrome (NMS)
Definition	Movement disorders caused by antipsychotic medications.	A rare but life-threatening reaction to antipsychotic drugs.
Onset	Gradual (usually within hours to days of starting treatment, may be seen months to years of drug intake).	Sudden (usually within days to weeks of starting treatment).
Symptoms	-Acute dystonia (muscle spasms) - Akathisia (restlessness) - Parkinsonism (rigidity, bradykinesia) - Tardive dyskinesia (involuntary movements, usually long-term)	- Hyperthermia (high fever) - Muscle rigidity (lead-pipe rigidity) - Autonomic dysregulation (tachycardia, high blood pressure, diaphoresis) - Altered mental status (delirium, confusion)
Etiology	Dopamine receptor blockade (usually from antipsychotics(FGA>SGA)).	Dopamine blockade, usually associated with high-potency antipsychotics .
Treatment	- Anticholinergic medications (e.g., benztropine) - Antihistamines (e.g., diphenhydramine) - Dose adjustment or change of antipsychotic	- Immediate discontinuation of antipsychotic drug - Supportive care (hydration, cooling) - Medications (e.g., dantrolene, bromocriptine) - Intensive monitoring
Severity	Generally non-life-threatening but can affect quality of life.	Life-threatening , requires urgent medical attention.
Management Focus	Symptom control and prevention of progression to tardive dyskinesia.	Emergency intervention to stabilize the patient and manage symptoms.
Duration	Chronic if not treated, but can resolve with treatment.	Can be fatal if not treated promptly, but may resolve with appropriate care.
Risk Factors	- Higher doses of antipsychotics - High-potency antipsychotics -Older age	- Recent introduction or increase in dose of antipsychotics - High-potency antipsychotics - History of NMS or similar reactions
Prognosis	Good prognosis with treatment, but can be chronic if untreated (especially tardive dyskinesia).	Poor without prompt treatment; better with early detection and treatment.

- ***Neuroleptic malignant syndrome (NMS)***

- potentially fatal idiosyncratic complication
- Rigidity, hyperpyrexia, delirium/confusion
- ***extreme EPS reaction*** resulting from excessive antidopaminergic activity.
- Precipitating factors include abrupt withdrawal of DA agonists (L-dopa) or metoclopramide, or rapid ↑ in the dose of neuroleptics.

- **EPS** are not fatal & are best treated with anticholinergics or antihistamines

- ***Tardive dyskinesia*** is difficult to manage, if symptoms appear at doses where the neuroleptic is therapeutically effective.

- Drug discontinuation and
- Substitution with an atypical neuroleptic, such as clozapine
- BZD, valbenazine

Clinical management of acute overdose

- As with tricyclic toxicity, cardiovascular and respiratory adverse effects with antipsychotics are monitored for 12 h, up to 6 days.
- Treatment for reversing neuroleptic induced hypotension and quinidine-like effects, especially ventricular dysrhythmias, is similarly managed with *NaHCO₃* & *lidocaine*.
- Convulsions or hyperactivity is controlled with *diazepam* or pentobarbital.

3. Acetaminophen, Salicylates (NSAIDs)

Acetaminophen

- Also called N-Acetyl-Para aminophenol, APAP, Paracetamol
- APAP is recommended as an analgesic/antipyretic
 - in the presence of aspirin *allergy*
 - in blood coagulation disorders & in patients(pt) who take oral anticoagulants
 - in *children*, in pt who have *upper GI disease*
 - in variety of arthritic and rheumatic conditions (musculoskeletal disorders, headache, & other minor pain)
 - for the management of *fever* associated with bacterial & viral infections.

Metabolism and Mechanism of toxicity

Metabolism

- Occurs via several pathways in the liver
 - 55% by glucuronidation
 - 35% by sulfation
 - 2% excreted unchanged in the urine
 - 5-10 % biotransformed by Cyp450 enzyme system
- produces reactive intermediate, N-acetyl-*p*-benzoquinone imine (NAPQI) metabolite.
- Further conjugation by cellular glutathione results in the production of mercapturic acid and cysteine conjugates

- With chronic use or with large doses,
 - glucuronide and sulfate conjugation routes are saturated
 - glutathione stores are depleted.
- The cyp450 oxidase pathway to accumulate toxic NAPQI metabolite.
- Binding of NAPQI to hepatocyte membranes and sulfhydryl proteins accounts for the hepatotoxic necrosis.

Signs and symptoms of acute toxicity

- ***Factors involved in predicting hepatotoxicity***
 - total quantity ingested
 - time from ingestion to treatment
 - age of the patient
 - alcoholism
 - enzyme inducing medications
- ***Toxic dose***
 - In adults, threshold for liver damage is 150 to 250 mg/kg

Clinical management of acute overdose

- Activated charcoal is beneficial within 1 to 2 hr post ingestion.
- An acetaminophen level obtained 4 hr later determines follow-up treatment with an antidote-NAC (so take blood level at the 4th hr)
- Plot level on Rumack-Matthews nomogram
 - This nomogram predicts clinical outcome and
 - guides use *therapeutic intervention or not*
 - 200 µg/mL at 4 hours is possibly toxic
 - *therapeutic “normal” values...*

Rumack-Mathew nomogram for acetaminophen poisoning

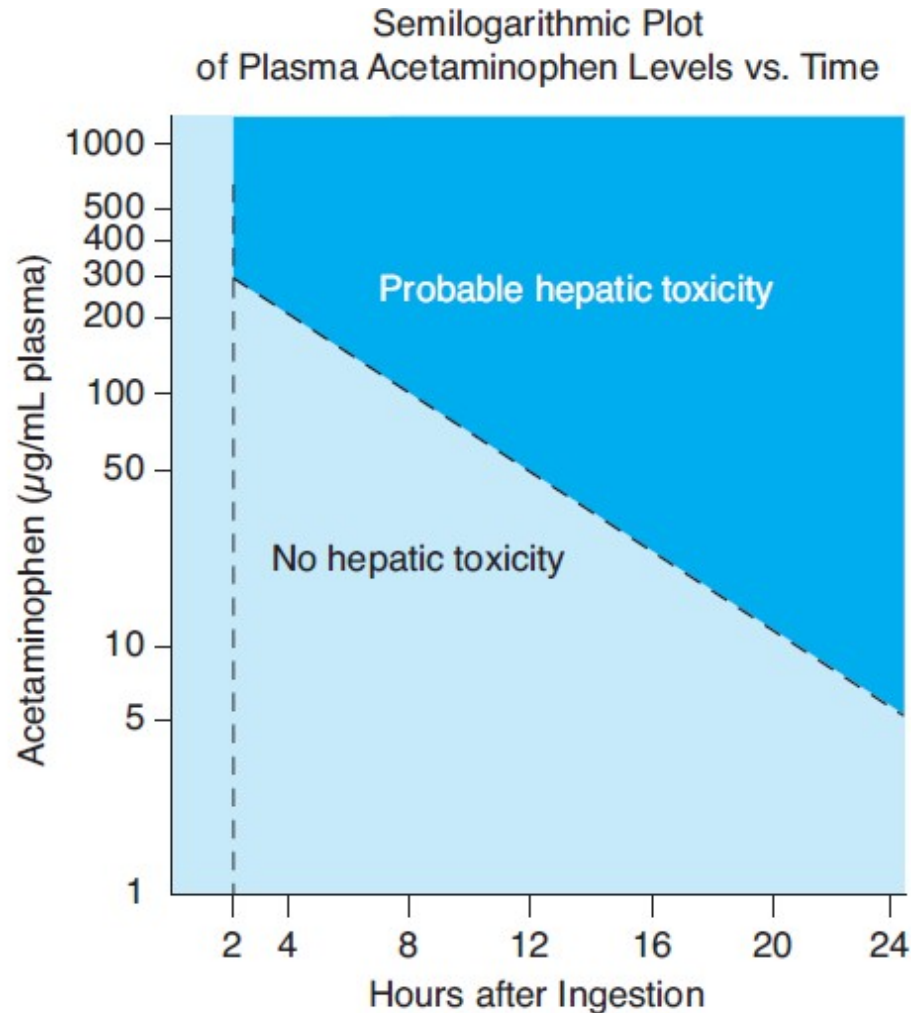


Figure 32-1. Rumack-Mathew nomogram for acetaminophen poisoning. (From Rumack and Mathew, 1975, with permission. Copyright American Academy of Pediatrics, 1975.)

Salicylates

❑ Acetylsalicylic acid (ASA or aspirin)

- aspirin is common in OTC oral preparations
 - Analgesic
 - Anti inflammatory
 - Antipyretic
- They are potential sources of toxicity (especially children, febrile)



Pharmacokinetics

- Rapidly absorbed in the stomach
 - Reach peak levels in 15-60 minutes
- 90% bound to *albumin* in the blood at a dose of 10 mg/dL
- 90% metabolized in the liver, 10% unchanged
- Metabolites & unchanged drug are filtered & secreted by the kidneys
- Hepatic enzymes become saturated and elimination follows
zero-order kinetics

Signs & Symptoms

- The earliest signs and symptoms of toxicity include nausea, vomiting, diaphoresis, and tinnitus with or without hearing loss
- Other CNS presentations include vertigo, hyperventilation, hyperactivity, agitation, delirium, hallucinations which are usually followed by convulsions, lethargy and stupor
- Renal failure, Cerebral edema, Cardiac arrhythmias.
- A marked elevation in temperature is a sign of severe toxicity

Pathophysiology

■ Acid-Base

- Salicylates clearly stimulate the respiratory center in the brainstem inducing *hyperventilation...respiratory alkalosis*
- Salicylates are also *weak acids* & can replace plasma bicarbonate coupled with impaired renal function (due to ASA toxicity) leading to accumulation of sulfuric & phosphoric acids
- Salicylates also uncouple oxidative phosphorylation that leads to accumulation of pyruvic & lactic acids & generates large amounts of heat
- The end result is *metabolic acidosis*

Salicylate poisoning indicators:

- **Whole blood glucose and electrolyte determination**
 - Shows ↓ bicarbonate and ↓ glucose & other electrolyte derangement
- **Whole blood ABG analysis**
 - Shows characteristic **acid-base disturbance** of salicylate toxicity

Management

- Within the initial presentation the use of gastric decontamination (**activated charcoal**) has shown to reduce the amount of active salicylate by 50-80%
 - Multiple doses of activated charcoal appears to be superior to single doses
- **Fluid replacement** is very important in the management of salicylate toxicity
 - Toxicity can induce major fluid losses through tachypnea, vomiting, hypermetabolic state, and insensible perspiration
- **Electrolyte replacement**

- The most important management is through urine alkalization
 - Alkalization with NaHCO_3 results in enhanced excretion of ionized acid form of salicylate
 - Alkalizing the urine from a pH of 5 to 8 increases renal clearance
 - Urine pH must be maintained at 7.5-8.0 and hypokalemia must be corrected
 - *Hypokalemia* is a common complication due to alkalemia, K^+ go to cells in exchange for H^+ ions
 - Calcium should also be measured as decreases are a complication of bicarbonate therapy

4. Steroids



ANABOLIC-ANDROGENIC STEROIDS (AAS)

- *Androgens* are hormones that stimulate the devt. & activity of accessory *male sex organs* (1^o Xs), & encourage the devt. of *male sex Xs* (2^o sex Xs).
- *Anabolic steroids* are **synthetic** derivatives of the androgens that **promote skeletal muscle growth (anabolic effect)** and devt. of male sex Xs (androgenic effect).

- Indicated for a variety of endocrine disorders
 - treatment of *hypogonadism*,
 - *delayed* puberty in males
 - metastatic cancer in females.
- Androgens stimulate adolescent growth and precipitate termination of linear growth by fusion of skeletal epiphyseal growth centers.
- Exogenous androgen admn. inhibits endogenous testosterone release through a pituitary LH feedback pathway.
- Anabolic steroids are closely related to testosterone and have high anabolic, low androgenic properties (*Nandrolone, oxandrolone*)

- Promote tissue-building and reverse tissue-depleting processes.
- Cause *serious disturbances of growth* and sexual devt. in children and suppress gonadotropic functions of the pituitary.
- Virilization is the most common undesirable side effect, especially in women and prepubertal males.



- In females: **virilization**

- clitoral enlargement.
- hirsutism
- deepening of the voice.
- Menstrual irregularities, male pattern baldness, acne, and changes in libido are reversible.

- In children and teenagers:
 - premature epiphyseal maturation and closure, without compensatory gain in linear growth.
 - disproportionate advancement in bone maturation and *termination of the growth* process.
- AAS are contraindicated during *pregnancy* (category **X**) because of potential fetal masculinization.
- In males
 - Testicles shrink
 - Breasts become larger
 - Low Sperm count
 - Infertility
 - Depression & aggression



Potential Negative Side Effects



Treatment of steroid use



- Psychological Support and Treatment – e.g Cognitive-behavioral therapy (*CBT*) and joining a support group to manage the growing problem of steroid abuse.
- Temporary Use of Testosterone Replacement
- Gradual Tapering vs. Abrupt Discontinuation...
- Medication management for co-occurring disorders like depression or anxiety (antidepressant)
- Depression is the most dangerous of the *withdrawal* symptoms, because of its association with suicidal tendencies.

5. Cardiovascular Drugs

DIGITALIS GLYCOSIDES

- DG poisoning may be acute or chronic
- Acute poisoning
 - suicidal ingestion of a large dose (adults).
 - unintentional ingestion in children.
- Chronic poisoning is more common, and typically develops in patients receiving DGs for therapeutic purposes.
- Manifestations of DG toxicity involve the CV, the CNS, and the GIT.



Conditions leading to Digoxin toxicity

- ESRD
 - ESRD prolongs half-life and reduces V_d
- Hypokalemia...
- Advanced age
- Cardiac diseases
 - Active ischemia, myocarditis, cardiomyopathy
- Quinidine, cyclosporine, verapamil, diltiazem, tetracycline, erythromycin, SSRI's
 - Most drugs raise levels by reducing *excretion*

Toxicity

- ↑d automaticity coupled with concomitant conduction delay
- Hyperkalemia
- No single dysrhythmia is always present
- Dysfunction of CNS, including delirium, fatigue, malaise, confusion, dizziness, abnormal dreams, blurred or yellow vision, and halos.
- Disturbances of color vision— a relatively specific sign of digoxin intoxication.
- GI disturbances may include anorexia, N, V, and abdominal pain.

Management

- Successful management depends on early recognition.
- Management varies
 - withdrawal of the medication,
 - general supportive care,
 - *Replace K^+ if hypokalemic*
 - proper treatment of the DG-induced cardiac *arrhythmias*
 - administration of digoxin specific Fab fragments (i.e., Digibind®) for life-threatening CV compromise (Ingested dose >10 mg (adult))

6. Opioid /morphine toxicities

- Narcotic- Morphine and related alkaloids
- Used to treat *pain, cough*, diarrhea, dyspnea
- Classified as natural (*morphine*) & synthetic (meperidine, methadone)
- **MoA-** interact with receptors in CNS & PNS
 - Mu (μ), kappa (κ), delta (δ) & sigma (σ) receptors



Morphine ...

- Common scenarios of poisoning
 - Accidental ingestion (children)
 - Therapeutic misadventures (acute or chronic accidental overdose)
 - Unintentional overdosing by drug abusers
 - Illicit drug users, sellers, or smugglers (intoxicated after ingesting drug packages or packaging them in body cavities)
 - Suicidal use

Signs & symptoms of poisoning

- The clinical presentation of the opioid toxidrome (triad) is Xd by *CNS depression* (coma), *miosis*, and *respiratory depression*
 - **Miosis**- by binding to μ receptors
 - **Coma**- all degrees of coma (drowsiness to complete absence of all reflexes)
 - **Respiratory depression** (bradypnea to total apnea)- due to decreased responsiveness of respiratory centers in brain stem to increases in PCO_2 .
 - depression of pontine and *medullary* centers regulating rhythmicity.
 - Death from *opioid poisoning* is usually due to **respiratory arrest**



Sign and symptoms...

- Dysphoria, stupor
- Noncardiac pulmonary edema (with severe overdose)
- Bradycardia, hypotension, & decreased GI motility.
 - Orthostatic hypotension (vasodilation due in part to histamine release)
 - Hypothermia, absent bowel sounds, constipation, abdominal distention, urinary retention (due to ↑ ADH secretion, ↓PSNS)

Management of opioid poisoning

- Supportive management
 - Maintenance of *vital functions*, including respiratory and cardiovascular integrity
 - An IV line & continuous cardiac and respiratory monitoring
 - Supplementary oxygen and assisted ventilation
- *Gastric lavage and induction of emesis* not routinely done
- *Activated charcoal* if came within 1hr of ingestion
- **Naloxone**- (IV in a dose of 0.03 mg/kg/dose) provides both therapeutic & diagnostic modality for opioid poisoning (reverses respiratory and CNS depression)

7. Sympathomimetics

Amphetamines and amphetamine like agents

- Are indirect agonists, either stimulate release, or block reuptake
- Have powerful CNS stimulant actions (**analeptic**)
- The toxicity is an extension of the pharmacological properties
- Indications are for the treatment of
 - Attention-deficit hyperactivity disorder (ADHD)
 - Narcolepsy
 - Short -term management of obesity?

Syndrome	Therapeutic Use	Euphoric Use	Toxic Overdose
<u>CNS Effects</u>	Increased alertness, wakefulness, focus	Euphoria, increased energy, heightened alertness	Agitation, restlessness, hyperactivity, tremors, confusion, seizures, delirium, paranoia, hallucinations
Mood	Improved mood, reduced fatigue	Heightened mood, feelings of invincibility or superiority	Anxiety, irritability, aggression, panic, psychosis
Motor Activity	Increased activity, improved task performance	Increased physical activity, excessive talking, hyperactivity	Tremors, hyperreflexia, clonus, involuntary movements
<u>Autonomic Function</u>	Mild increase in blood pressure and heart rate	Tachycardia, increased systolic blood pressure	Severe hypertension, arrhythmias, hyperthermia
Appetite	Decreased appetite (in some cases)	Further suppression of appetite	Extreme suppression of appetite, malnutrition risk
Sensory Perception	No significant effect on sensory perception	Heightened sensory perception (often distorted)	Hallucinations, auditory and visual distortions
Duration of Effects	Short duration (several hours)	Often short-lasting, but may extend as person becomes tolerant	Prolonged effects; symptoms can last for hours to days if overdose is severe
<u>Other Effects</u>	None, or minor subjective discomfort (e.g., dry mouth)	Increased talkativeness, sexual desire	Risk of stroke, cardiac arrhythmia, seizures, death due to overdose

Amphetamine toxicity

- ***Treatment modalities*** include
 - Supportive care -ABC & monitor V/S
 - *Activated charcoal* if early
 - Regulating cardiovascular effects, and
 - BZD for aggression and seizure
 - *Antiarrhythmic* for arrhythmia, monitor with EKG
 - *Cooling* measures- for hyperthermia
 - Therapeutic intervention for the toxic psychosis syndrome (*haloperidol, chlorpromazine*) for refractory cases of psychosis

Cocaine

- Indirect acting (*Dopamine* Reuptake Blockade, others *NE*, serotonin reuptake also be blocked)
- Has stimulant as well as anesthetic properties
- Displays reversible CNS and peripheral actions
- Increased *sympathetic* activity (HTN) due to high *NE* in synapse
- Production of *euphoria*, *psychosis*, *addiction* is due to inhibition of DA reuptake in central synapses
- Blocks nerve conduction by inhibiting neuronal sodium permeability (LA)
- Possesses local vasoconstrictor actions secondary to inhibition of NE reuptake...nasal mucosa damage, bleeding, perforation

Signs and symptoms of Cocaine toxicity

Cardiovascular Symptoms:

- Severe hypertension
- Tachycardia or arrhythmias
- Chest pain (which can be a sign of myocardial infarction or heart attack)

CNS Symptoms:

- Agitation or restlessness
- Severe anxiety or panic attacks
- Psychosis/hallucinations (auditory and visual)
- Delirium or Seizures

Respiratory Symptoms:

- Hyperventilation (rapid breathing)
- Respiratory distress (shortness of breath)

Other Physical Symptoms:

- Hyperthermia
- Sweating, flushing, or skin pallor
- Dilated pupils
- muscle spasms
- Loss of consciousness or coma
- Acute renal failure
- production of acute dyskinesia (*crack dancing*)
- Increased risk of spontaneous abortions

clinical management of acute overdose

- Maintenance of airway, breathing, and circulation (ABCs) is the priority.
- **Activated charcoal**
- **BZDs** (e.g diazepam) for agitation, seizure
- **IV Phenobarbital**: to control seizures if BZDs are inadequate.
- **Alpha-blockers** (e.g phentolamine) for HTN and $\uparrow$ HR...but avoid Beta-blockers
- **antiarrhythmic therapy**-ventricular arrhythmias or atrial fibrillation
- **Cooling measures**: for hyperthermia for $>40^{\circ}\text{C}$
- **Antipsychotics** for refractory cases of psychosis

8. Others

❖ Xanthine derivatives

- enhance cardiac muscle contraction
- induce coronary vasodilation but cerebral constriction
- bronchodilation
- promote diuresis

☐ Caffeine

- stimulates cerebral activity
- wakefulness and mental alertness, ↓fatigue (C>T)
- skeletal and cardiac muscle contraction
- ↑general basal metabolic rate
- Migraine??
- weight loss (with many fat burners & supplements)

Caffeine ...

- treatment of apnea in premature infants (caffeine citrate also theophylline)
- Signs and symptoms
- LD of caffeine in humans is 10 g or more.
- dysrhythmias or strychnine-like seizures (10 mg/kg).
- More likely adverse effects of excessive caffeine intake are
 - CNS stimulation (insomnia, restlessness, sensory disturbances, and delirium)
 - ↑skeletal muscle tension
 - Tachycardia,
 - premature ventricular contractions (PVCs)
 - diarrhea, development of peptic ulcers and GI bleeding

clinical management of caffeine toxicity

- Myocardial tachyarrhythmias and development of seizures should be monitored in patients after an acute ingestion of *1g* or more of caffeine.
- *Myocardial problem*: short-acting β -blocker, such as esmolol,
- *Seizures*: short-acting BZD, such as midazolam

□ Theophylline

- less central stimulation at normal dose
- significant bronchial smooth muscle relaxation
- treatment of bronchial asthma and other respiratory disorders

signs and symptoms of theophylline toxicity

- Although its toxicity is more acute and more common; chronic toxicity, however, is unlikely.
- Therapeutic blood levels: 10 to 20 $\mu\text{g/ml}$. Toxic >20
- Seizures occur: 25 to 40 $\mu\text{g/ml}$.
- Rapid i.v. admin is associated with headache, hypotension, dizziness, restlessness, agitation, and

❖ Nicotine

- Stimulates nicotinic receptors
- Signs and symptoms occur 1 to 2 h after exposure and include
 - Initial generalized stimulation (SNS & PSNS)
 - Followed by predominance of parasympathetic overtone (salivation, lacrimation, urination, defecation, vomiting). HTN, ↑HR, tremor (SNS)
- Continued exposure progresses to (desensitization)
 - muscular weakness
 - bradycardia
 - hypotension
 - dyspnea
 - convulsions and respiratory paralysis, coma

- **Treatment of poisoning**

- primarily symptomatic(ABC, Diazepam, atropine)
- involves washing of the agent from the skin with water
- **activated charcoal** if intervention starts soon after ingestion
- induction of emesis or gastric lavage not routinely done
- maintenance of vital signs
- **BZDs**
- **Atropine-** for bradycardia

9. Strychnine

- Exhibits a narrow therapeutic index
- No favorable medicinal value (poison)
- Used as a rodenticide.
- Extremely toxic as a central stimulant.
- Fatal poisoning in adults: 30 to 100 mg
- In children, from accidental ingestion of rodenticide (15-mg can be a lethal dose).
- Inhibits the postsynaptic receptor for glycine
- muscle spasm, Hyperreflexia, seizures, Sweating, tachycardia, hypertension, respiratory failure

● Treatment:

- maintaining ventilation
- *Activated charcoal*-is indicated when the cpd is suspected to be in the stomach contents
- *Gastric lavage*...not routinely done
- prevent convulsions (admin of an anticonvulsant)
- skeletal muscle relaxant (NM blocker- e.g pancuronium)

8. Hallucinogenic Agents

LYSERGIC ACID DIETHYLAMIDE (LSD)

- experimental value as a model for psychosis.
- Within minutes of ingestion, an effective dose produces hallucinogenic sensations that last between 6 and 24 h.
- produces significant pyramidal and extrapyramidal effects as a result of interaction with central (5-HT/serotonin) receptors.
- it is non-selective serotonin rec agonist
- activates 5-HT_{1A} receptors on raphe nucleus cell bodies and acts as a partial or full agonist at 5-HT_{2A/2C} receptors(hallucinogenic effect), it also act on 5-HT₅₋₇ rec

Hallucinogenic effects...

- Swift mood swings.
- **LSD** *trip*: distortion in the realization of time & alteration of sensory perception, especially visual, auditory, tactile, olfactory & gustatory.
- Altered perception of color.
- Other distortions of reality include synesthesia-e.g auditory stimulus leads to visual perception
- Normal taste, smell & touch sensations are often strange, distorted, or uninterpretable.

Sign and symptoms of acute toxicity

- Sympathetic stimulation
 - mydriasis
 - Tachycardia & hypertension
 - hyperthermia
- Panic reactions and loss of psychological control of the immediate environment.
- *Flashbacks* occur in 1 of 20 users.
 - recurrence of the LSD experience in the absence of ingestion
 - somatic and emotional hallucinations
 - development of personality changes
 - prolonged psychosis, neurosis, and depression
 - flashbacks are unpredictable(variety, intensity, & pattern do not rely on *dose or frequency* of use).

Clinical management

- Reassurance that a bad trip is due to the effects of the drug and is not permanent.
- BZDs for agitation, seizure.
- Management of symptoms of sympathetic stimulation, especially vital signs(tachycardia) and hyperglycemia,
- Cool the patient if hyperthermic
- Antipsychotics are not recommended because of the lowering of seizure thresholds.

Marijuana

- *Dronabinol*, a synthetic form of δ^9 -THC (5&10 mg capsules)
- treatment of N & V associated with cancer chemotherapy.
- appetite stimulant for anorexia with weight loss in AIDS patients & others
- treatment of glaucoma & epilepsy.
- Nontherapeutic effects of dronabinol are identical to those of marijuana & other centrally active cannabinoids.

Effects

- Smoking marijuana or oral ingestion of cannabinoids
 - has complex central *sympathomimetic* effects, mediated in part by cannabinoid receptors.
 - mood changes, ranging from euphoria, depression, paranoia, & anxiety.
 - most prominent effects are relaxation and sedation.
 - loss of goal-oriented drive and short-term memory, and a vague sense of time (temporal disintegration).
 - spontaneous laughter, hallucinations, & delusions (with higher doses)

Effects

- dry mouth and red eyes
- stimulation of appetite
- muscular incoordination(ataxia)
- ↓testosterone levels
- urinary retention
- ↑ in HR
- ↓ IOP

Acute toxicity and clinical management

- toxic effects are an extension of the pharmacological effects
- serious toxicity is uncommon
- psychosis and dangers related to poor judgment are complications
- psychiatric reassurance and supportive care
- Activated charcoal if early (1hr)
- psychosis is transient and manageable with BZDs
- Antiemetics- for persistent nausea & vomiting
- Monitor vital signs